TLS PROPOSAL

NOT FOR FABRICATION

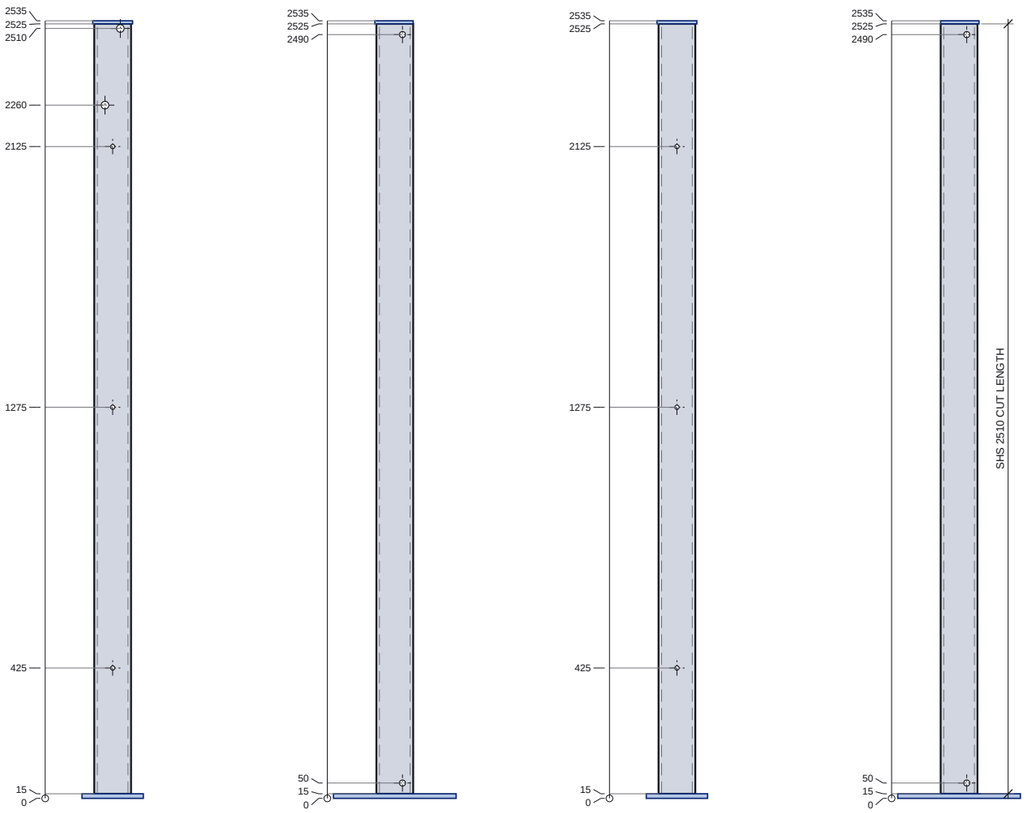
FOR ENGINEERS REVIEW

until the engineer has reviewed + signed off

PC1 FACE ELEVATIONS

Scale 1:20 | heights above the underside of the baseplate (-125)

Base underside -125 = height 0; top of cap +2410 = height 2535



N FACE — from N: E left, W right

E FACE — from E: S left, N right

S FACE — from S: W left, E right

W FACE — from W: N left, S right

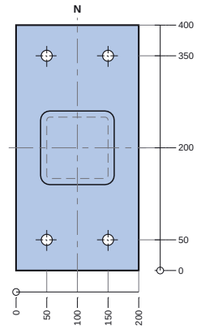
Plates dashed = behind the post in that view. Holes: see the schedule below.

HOLE SCHEDULE

No.	FACE	HEIGHT	POSITION	HOLE	PURPOSE
H1	E	2490	25 N	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H2	W	2490	25 S	Ø20	Ø20 galvanising vent / drain (top), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H3	E	50	25 N	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H4	W	50	25 S	Ø20	Ø20 galvanising vent / drain (bottom), HDG drive-in plug after galvanising [TLS detail - engineer to confirm]
H5	N	2260	25 E	Ø25	Ø25 through the inner face inside the RHS footprint: links + drains the RHS and SHS hollows for galvanising (drill before welding) [TLS detail - engineer to confirm]
H6	N	2510	25 W	Ø25	Ø25 through the inner face inside the RHS footprint: links + drains the RHS and SHS hollows for galvanising (drill before welding) [TLS detail - engineer to confirm]
H7	through-NS	425	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]
H8	through-NS	1275	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]
H9	through-NS	2125	on CL	Ø14	M12x160 through-bolt for the jamb plate (snug tight) [TLS detail - engineer to confirm]

BPPC1 BASEPLATE PLAN 1:10

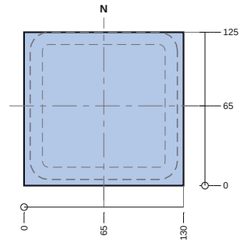
from above, N up



PL 400 x 200 x 15 S275 (9.30 kg)
4 no. Ø18 holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 8mm FW all round
TLS detail - engineer to confirm

KPPC1 CAP PLATE PLAN 1:5

from above, N up



PL 130 x 125 x 10 S275 (1.28 kg)
no holes; dims from the SW corner; post CL dash-dot
Weld to the SHS: 6mm FW on the three oversailing sides; inner face flush, sealed by the RHS end weld
ECCENTRIC: extends 65 E / 65 W / 60 N / 65 S of the post CL
TLS detail - engineer to confirm

CUT LIST

MARK	ITEM	GRADE	LENGTH	QTY	KG
PC1	SHS120x120x10	S355J2H	2510	1	84.6
BPPC1	PL 400x200x15	S275		1	9.30
KPPC1	PL 130x125x10	S275		1	1.28
TOTAL					**95.2 kg**

CONNECTIONS

Detail GP1 knee: SB4 end A/B on PC1/PC2 - Shop-welded knee (R4): RHS square-cut ends welded all round to the post inner faces, 8 FW; BT1 ends 6 FW to the post faces. Frame galvanised as one piece (galvaniser to confirm the bath + dip; temporary bolted spreader between the bases until set; post centres +/-3 after galvanising). FOR ENGINEERS REVIEW

NOTES

1. **FOR APPROVAL - NOT FOR FABRICATION until the engineer has approved the TLS details and returned the steel drawings.**
2. ALL STEELWORK SUBJECT TO SITE MEASURE BEFORE FABRICATION.
3. BS EN 1090-2 EXC2, CE marked. SHS S355J2H (EN 10210-1), plates S275JR.
4. Holes: normal clearance (EN 1090-2 Table 11): M12 = 14, M16 = 18. Drilled or punched full size, deburred.
5. Running dimensions from end A / base underside (o = datum). Levels above FFL (model 0).
6. Coating: Hot-dip galvanised to BS EN ISO 1461 (thickness to Table 3); T-wash + paint where exposed (architect).
7. Vent / drain holes: plug after galvanising.
8. Holes / plates marked TLS detail are not on the engineer's drawings: TLS detail - engineer to confirm before fabrication.
9. GP1 - TLS PROPOSAL R4 FOR ENGINEERS REVIEW (replaces S-0010 UB203x133x30); calc on the GP1 sheet
10. PC1 and PC2 steel are handed pairs (inner-face holes face each other): jamb plate faces differ, see the timber schedule.
11. GP1 R4 - PRELIMINARY, FOR ENGINEER REVIEW (TLS proposal, not the engineer's design). Replaces S-0010's UB203x133x30: carries the 5300 door opening AND ~0.5 m of stone (main-house stone eaves).
12. LOADS (char.): trusses S-0012 DL 15 / IL 10 kN each end -> 5.4 / 3.6 kN/m; stone 0.5x0.1x23 = 1.15 kN/m; boot + timber + self weight 1.0 kN/m; door rails + operator 2.0 kN mid-span; wind 12 kN SLS / 18 kN ULS at the eaves.
13. RHS 300x100x12.5: I 9103 cm4, Wpl 806 cm3. ULS M 61 kNm vs Mpl 286 kNm. Deflection (SLS incl. door, pinned, L 5420) 6.9 mm = L/787 < 8 mm (client limit).
14. SHS 120x120x10 to TOS: N 44 kN + knee M 29.7 kNm vs Mpl 62 kNm; sway 6.8 mm = h/326.
15. KNEE (R4): RHS ends welded all round to the post inner faces, 8 FW: couple 29.7 / 0.288 = 103 kN per RHS wall vs 136 kN; post face (b1/b0 0.83) ~390 kN; torsion from the boot 0.84 kNm per knee.
16. BASE: 400x200x15, 54 clear of the stone leaf; N 44 kN, H 9 kN per base (ULS); pinned base assumed for sway. Boot toe ~0.22 kNm/m -> 13 MPa in the 10 plate.
17. FOR THE ENGINEER: loads, wind share, knee welds + post face (stiffeners in the SHS?), base fixity, padstones on the garage return foundations (min 500 x 300), cavity tray / thermal bridge at the boot, 30 min fire protection (S-0011 9.02).

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

PC1 SHS120X120X10 POST FABRICATION

GP1 portal post to TOS; the RHS frames into its inner face (R4)

SCALE	SIZE	DATE	DRAWN
1:20 / 1:5 / as noted	A3	11.10.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	P4-ST-PC1	C03

Do not scale. All dimensions in mm. Check all dimensions on site.